Operations Management

Session 4: Managing Process Variability

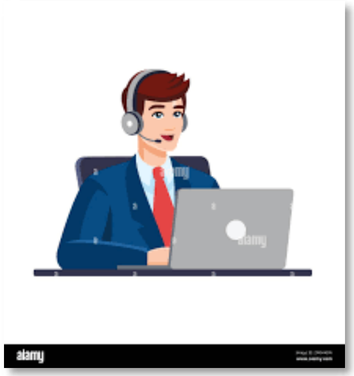
Outline:

- Measuring and Managing Variability
  - The variability–congestion relationship
  - The hidden dangers of high utilization
- Sof-Optics
  - Ops Quadrangle tactics for managing delays
  - Staffing and scheduling as queue control
- Queueing Fundamentals
  - G/G/c Formula
  - Capacity Pooling
  - When to add capacity vs manage demand
- The Psychology of Waiting

Prof: René Caldentey

CHICAGO BOOTH
The University of Chicago Booth School of Business

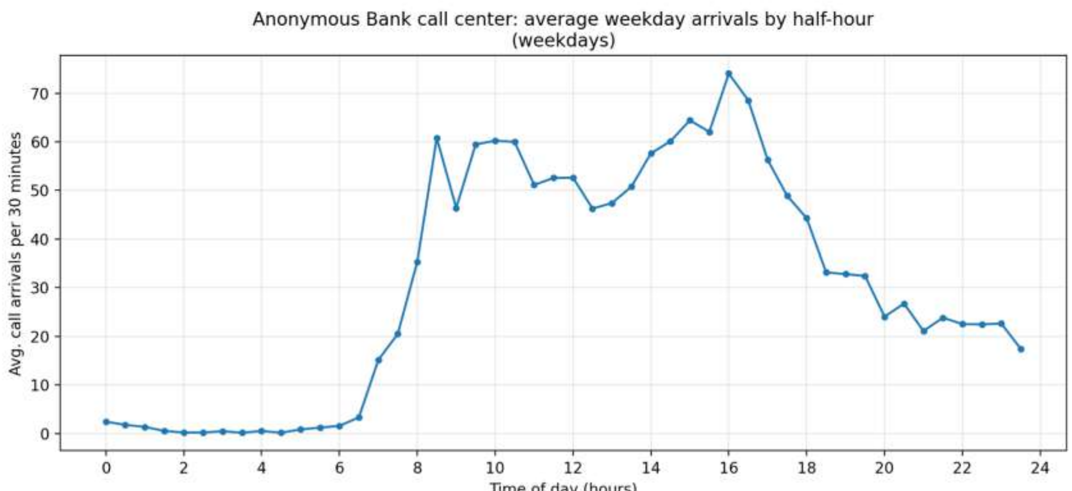
Question



Call Center

- On average, 1,600 calls per day (8-hour day)
- Each call takes 3 minutes on an average
- Each agent works for 8 hours every day.

Question: What is the optimal number of employees to have?
Answer: We should have 10 employees



Source: Guedj and Mandelbaum [Link]

NO!

- Unless calls arrive uniformly (exactly 1 call every 18 seconds) and every call takes exactly 3 mins., there will be a build-up of "waiting customers".
- In fact, if we have only 10 employees, a large percent of customers will hang up before being answered.

Process Variability

Process variability refers to the fluctuations and inconsistencies that occur during the execution of an operational process. It can impact various aspects of production or service delivery, including output quality, time, cost, and overall efficiency.

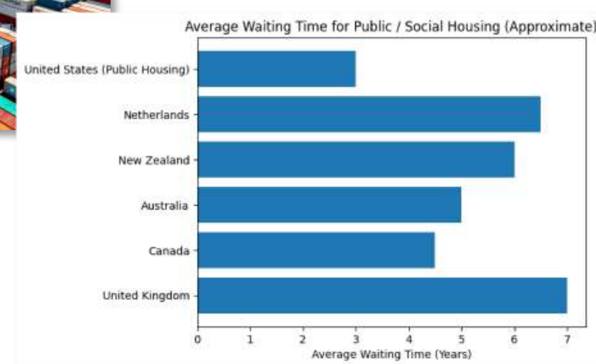
Variability can result from numerous factors:

- **Demand Variability**: Fluctuations in customer demand for goods or services due seasonality, sales promotions, economics factors, leading to inconsistent workload levels.
- **Processing Time Variability**: Variations in how long it takes to complete a task or process step. This could be due to differences in employee performance, machine speed, or tool effectiveness.
- **Input Variability**: Inconsistencies in the quality, type, or quantity of materials or components used in the process.
- **Machine Variability**: Unpredictability in machine performance due to wear and tear, maintenance needs, or breakdowns.
- **Human Variability**: Differences in how people perform tasks, are influenced by factors like skill levels, absenteeism, fatigue, or attention.
- **External Variability**: Uncontrollable external factors such as weather, supply chain disruptions, or regulatory changes that impact process performance.

Sources of variability	Ways to reduce
Arrival stream variation	Appointments, scheduling
Sales promotions	Everyday low pricing
Natural worker variation	Training, technology
Lunch breaks	Staggered lunches
Absenteeism	Temp help
Breakdowns, equip. failures	Maintenance, redundancy
Easy, hard jobs	Special procedures

We cannot eliminate all variability!

Variability & the Inevitability of Waiting



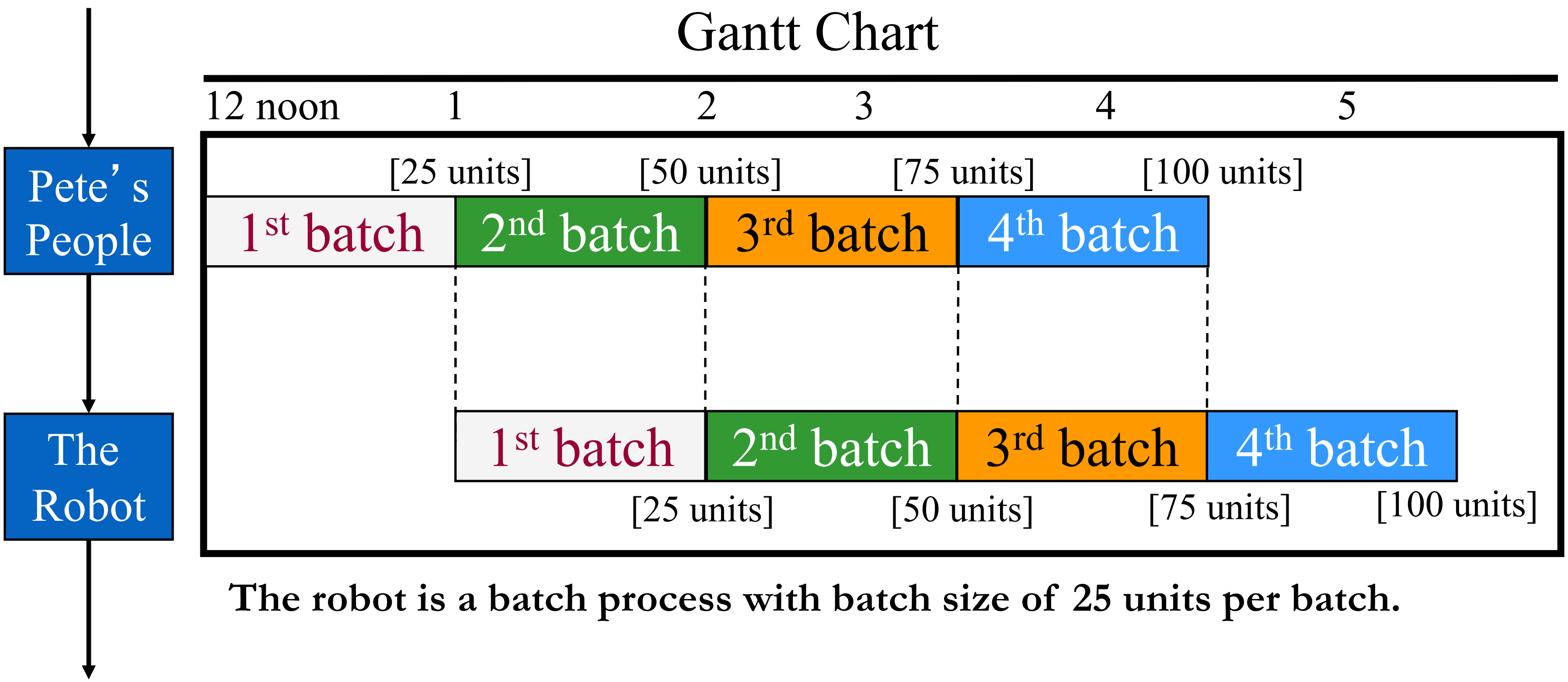
OECD, Social Housing: A Key Part of Past and Future Housing Policy

(NYT) Where the time goes:

- 6 Months: sitting at stoplights
- 1 year: looking for misplaced objects
- 4 years: doing housework
- **5 years: waiting in line**
- 6 years: eating

Variability: A Waste in Capacity

I. Ideal Situation: No variability

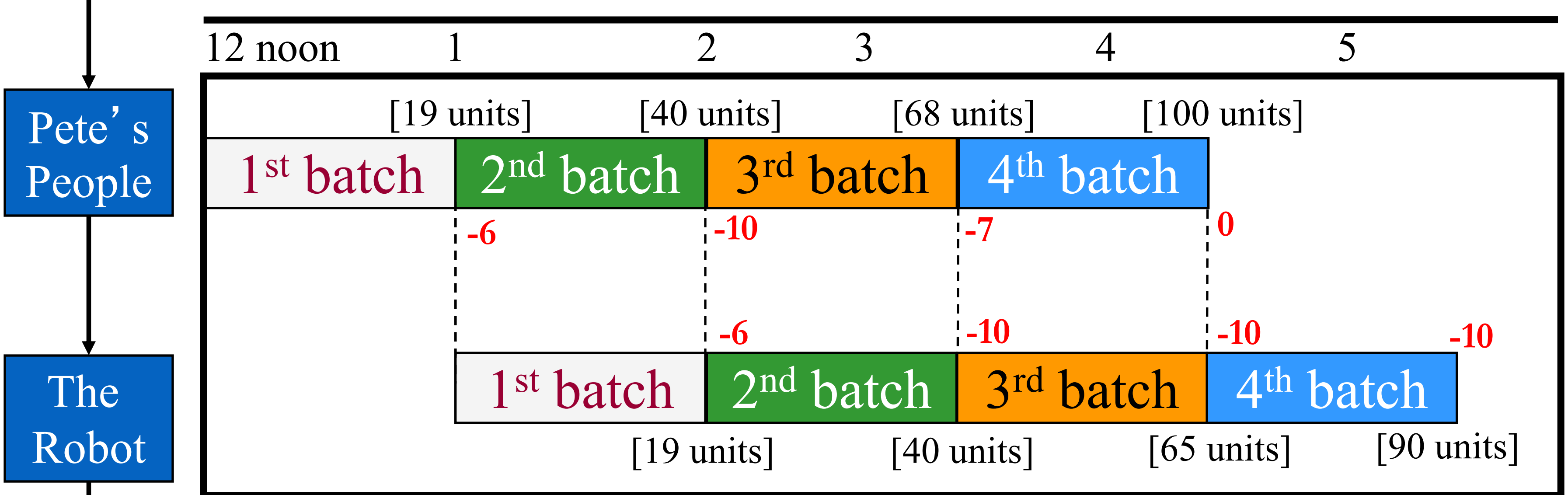


The robot is a batch process with batch size of 25 units per batch.

Variability: A Waste in Capacity

II. Real Situation: There is variability

Gantt Chart



The robot is a batch process with batch size of 25 units per batch.

Variability ***reduces*** the capacity utilization of the process

Variability: A Waste of Service Performance

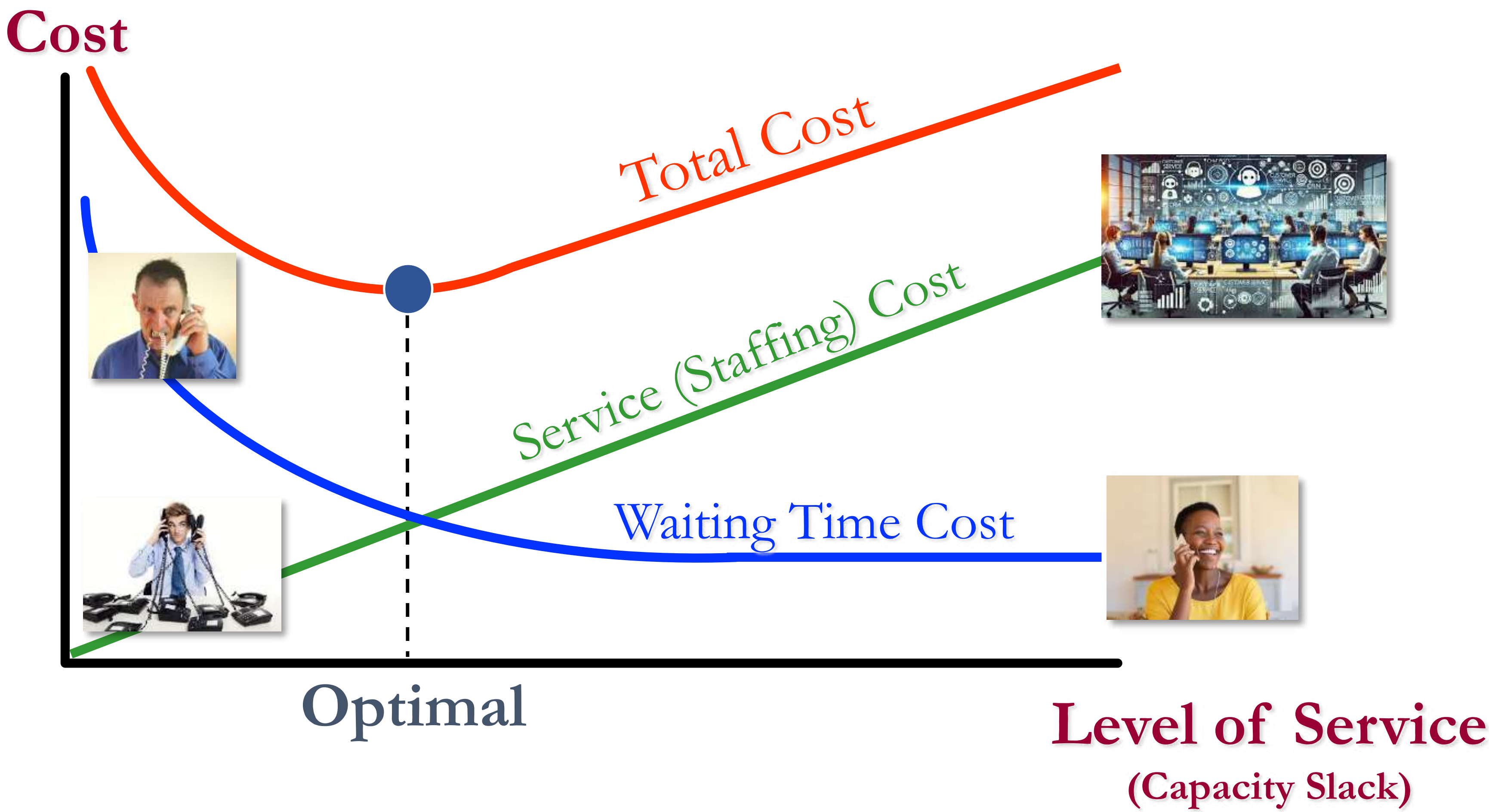
Industry	*Process*	*Avg. CT*	*Value-Add*
Life Insurance	New policy application	72 hours	7 minutes
Commercial bank	Consumer loan	24 hours	34 minutes
Hospital	Patient billing	10 days	3 hours
Motor vehicle equipment	Financial end-of-month close	11 days	5 hours
Airframe manufacture	Engineering change orders	21 days	1.75 days

Adapted from: Blackbrun: "Time Based Competition: White Collar Activities," Business Horizons, July-Aug. 1992

$$\text{CT Performance} = \frac{\text{Theoretical CT}}{\text{Avg. CT}}$$

$$\underset{\text{Hospital}}{\text{CT Performance}} = \frac{3 \text{ hrs}}{10 * 8 \text{ hrs}} = 3.75\%$$

A Two-Part Solution: Cost Tradeoffs



A Two-Part Solution: Process Variability

$$\text{Waiting Delays} \;\propto\; \text{Process Variability} \times \frac{1}{\text{Capacity Slack}}$$

Sof-Optics, Inc. (A)

Prof: René Caldentey

Contact Centers
(More than 300 billion calls annually globally)

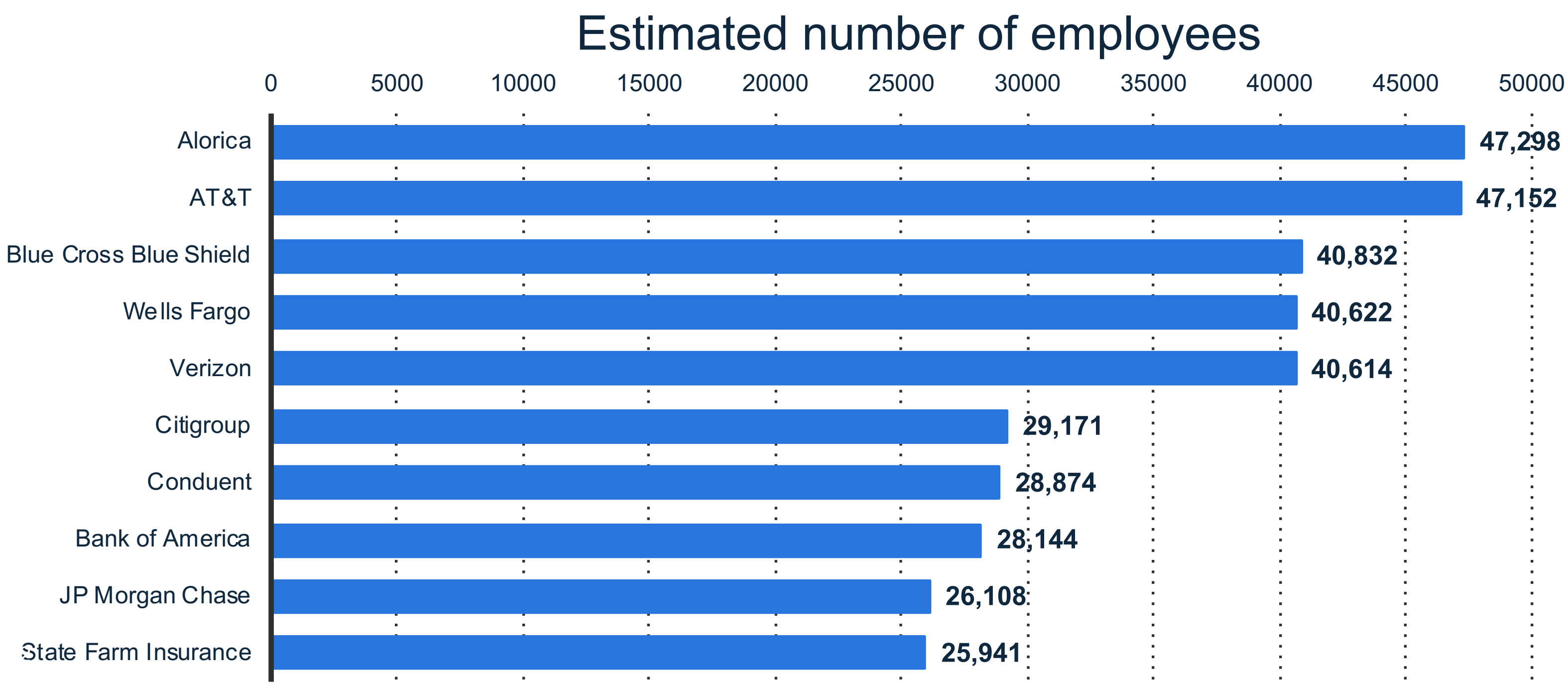
- 50% of customers prefer live phone support when contacting a company
- 67% of customers have hung up the phone out of frustration because they couldn't talk to a real person
- After a positive customer experience, 95% of Americas will use company again and 69% of them would recommend that company to others
- Following a negative customer experience, 58% of Americans would never use that company again
- 85% of customers are willing to pay more for a better customer experience
- The average response time for a call center agent is around 5 minutes.
- 81% of consumers: query resolved on first contact
- Close to 3 million contact center agents in the US
- Call centers experience an average of 40-60% absenteeism rate among their agents

Source: *10 Customer Service Stats and What They Mean for Your Contact Center*
Worldmetrics.org

Top 5 Industries Based on U.S. Call Center Employment

Rank	Industry	# of Employees	# of Call Center Facilities
1	Business Process Outsourcing	536,883	1,691
2	Financial Services	500,446	1,065
3	Insurance	358,508	775
4	Telecommunications	230,172	625
5	Healthcare	186,843	659

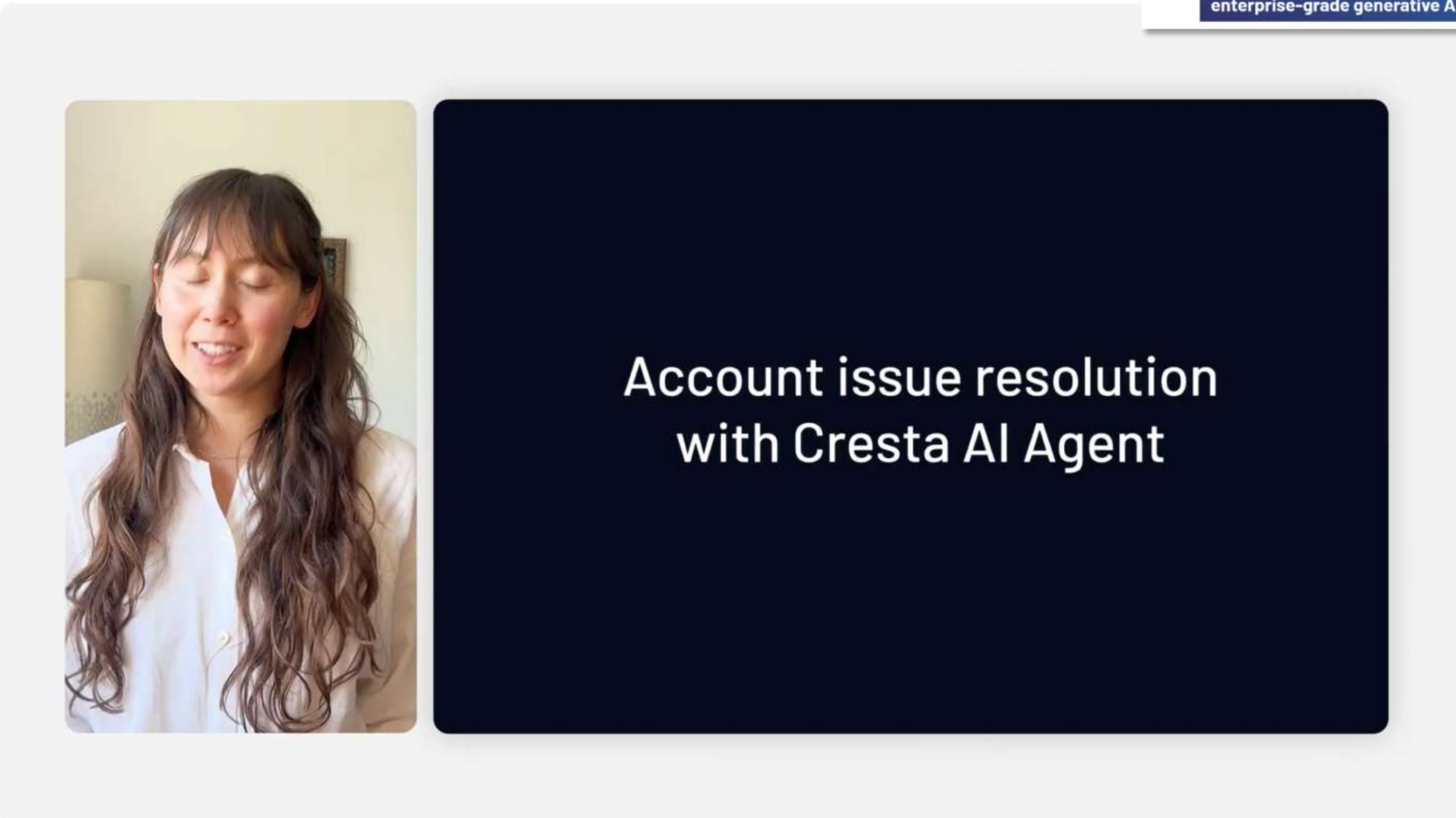
Source: Website (siteselectiongroup.com) Apr, 2021



Largest U.S. call centers by number of employees 2018

Source: Website (siteselectiongroup.com)

AI-Assisted Contact Center



Available on YouTube [Link]

AI-Assisted Content Moderation

7.9 million

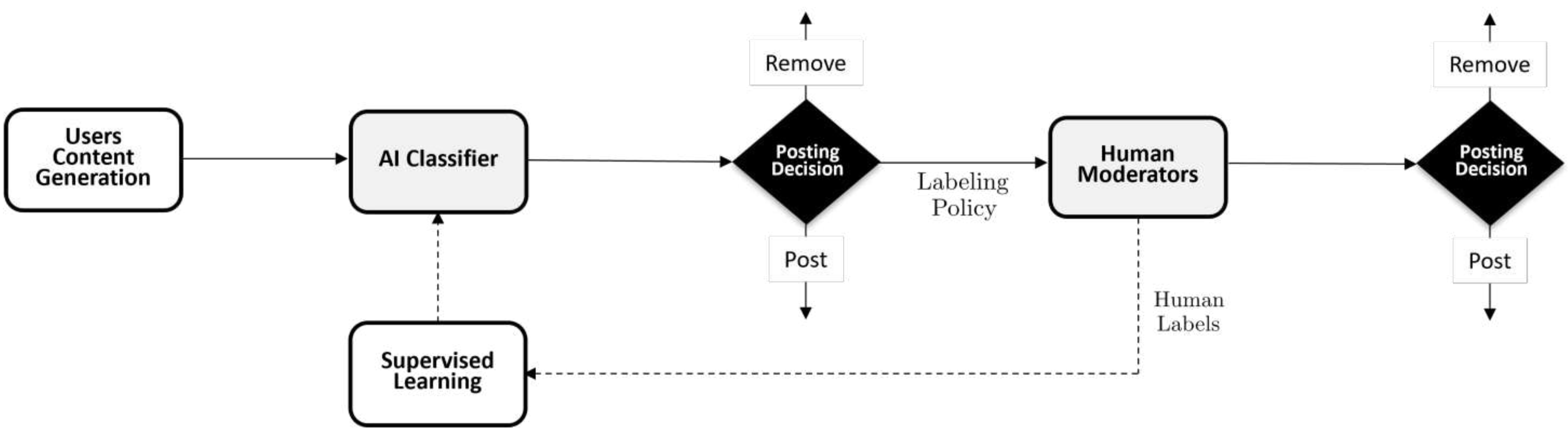
Content Actioned on Bullying and Harassment

65.8%

Actioned on Bullying and Harassment proactively

over 15k

Reviewers Review Content in over 50 Languages



Issues/Problems Facing S-O

Sof-Optics, Inc. (A)

<u>Problems</u>

- company not earning profits
- quality and yield problems
- customers face long waits for calls, or receive a busy signal
- lost sales due to busy signals and abandoned calls
- high stress, turnover and absenteeism among CSRs

<u>Causes</u>

- factory not operating at capacity, high operating costs relative to sales
- high inventory levels
- variability in call arrivals and CSR performance!
- only 12 WATS lines and 8 CSRs mean many waits and lost calls
- strenuous work and scheduling of CSRs creating turnover

Inventory Management at Sof-Optics

Figure B

Category	Sales Volume (per week)	# of Standard Lens Type	Batch Size (per Lens Type)	Reorder Pt. (month’s supply on hand)
A	50 or more lenses	16	500	1.5
B	20-49 lenses	3	250	2.0
C	Less than 20 lenses	21	200	2.5

Figure C Yield Data

	FY’78	FY’79 ′	FY’80	FY’81 (Est.)
Variable cost (per good button)	\$12.00	\$7.25	\$5.00	\$4.75
Average yield (button to lens)	29%	49%	64%	75%
Variable manufacturing cost (per good lens)	\$41.38	\$14.80	\$7.82	\$6.33
Annual volume (000 units)	36	104	211	400

Category A:

–) Avg. Monthly Demand $\geq$ 400 units

–) Reorder Point (ROP) $\geq 400 \times 1.5 = 600$ units

–) Production Batch Size (B) = 500 units

–) Avg. Inv. per Type = (ROP+B)/2 = 550 units

–) Avg. Inv. Category = $550 \times 16 = 8{,}800$ units

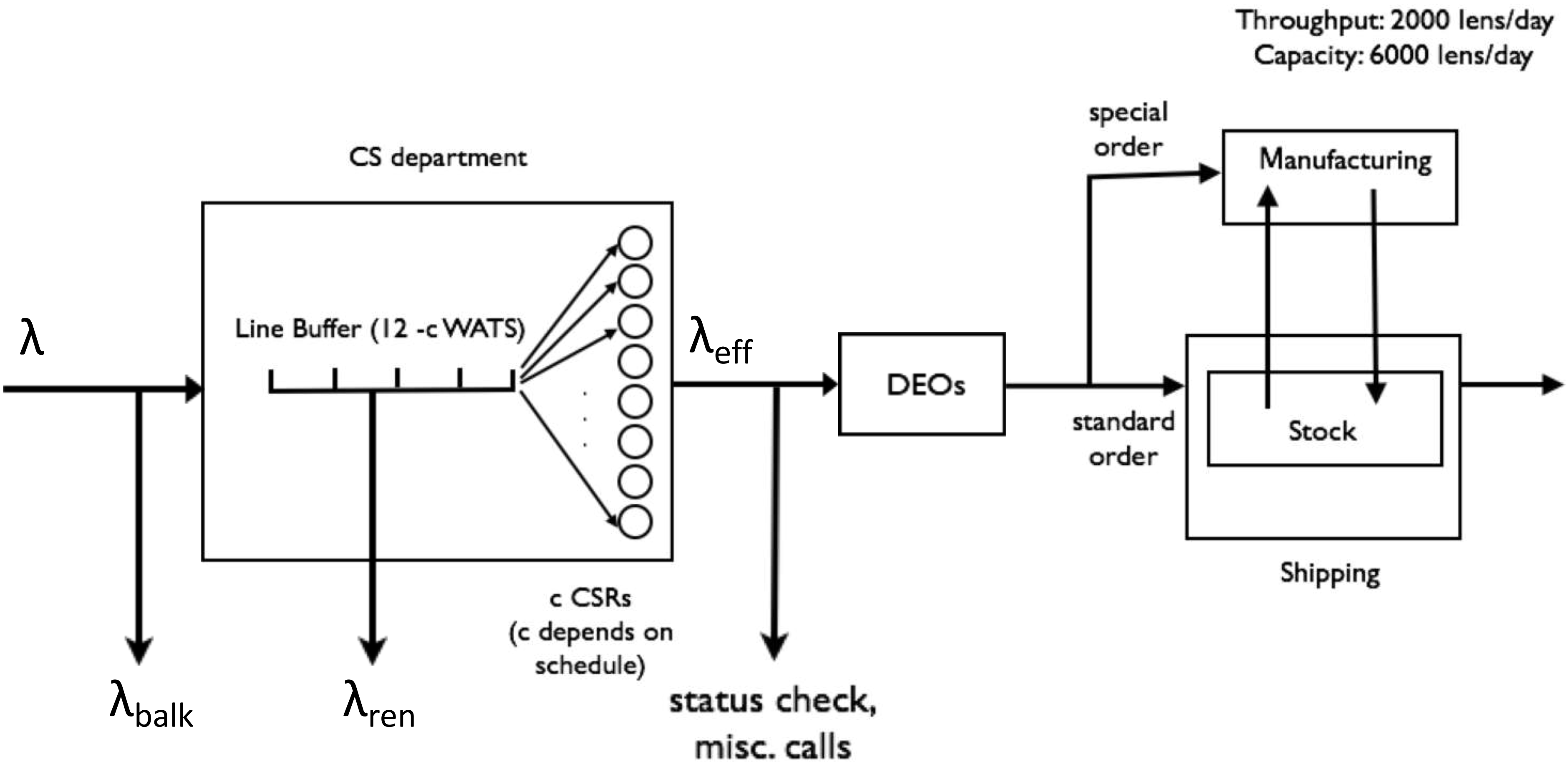
Category B: Avg. Inv. Category = 795 units

Category C: Avg. Inv. Category = 2,363 units

Total Avg. Inv. Categories A,B and C = 8,800+795+2,363 $\approx$ 12,000 units

Total Inv. Cost: Categories A,B and C $= 12,000 \times (\$5.00 + \$7.82) \approx \$154,000$

Process Flow Diagram



$$\boxed{\lambda = \lambda_{balk} + \lambda_{ren} + \lambda_{eff}}$$

Focus on the CS department!

What does a lost call cost Sof-Optics?

A simple first/cut:

Exhibit 1: Gross Mar = 10.5M (lens rev)-2.5M (lens cost) = $8M

Exhibit 8: (1164 calls/day)(250 days/yr) = 291 K calls/yr

therefore:

Margin/Call = (8 $M/yr) /(291 K calls/yr) = $ 27.5 /call

so,

$\underbrace{\text{(136.3 lost calls/day}}_{\text{Exhibit 8}}$)(250 days/yr)($ 27.5/call) ≈ $1 Million/yr

IS THIS ESTIMATE TOO HIGH OR TOO LOW?

How to recover some of these lost sales?

Cycle Time

- Reduce processing time:
  - during hold, ask for information, better scripting
- Reduce wait time (perception)
  - play pleasant music, distractions start service earlier, call back, etc.

Capacity

- **hire more CSRs**
- online, web, etc.
- pool CSRs and DEOs
- specialize CSRs

Throughput/Demand

- automatic status check
- quantity discounts,
- encourage batching
- influence call times, “early bird”
- proactive calling, online, web, etc.

Inventory

- expand consignment program
- prioritize calls (different classes of inventory)
- increase the number of WATS lines

What is the capacity of a CSR?

Exhibit 8 Call Statistics—Daily Average for October 21-23

Half Hour Ending	Number of Calls	Number Delayed	Average Delay (Seconds)	Number Abandoned[a]
6:30 A.M.	22.7	5.7	21.2	1.3
7:00	26.7	10.7	28.6	2.7
7:30	52.3	41.7	35.7	4.7
8:00	54.0	47.7	49.3	5.0
8:30	55.7	49.7	61.4	6.3
9:00	58.3	59.3	73.8	7.7
9:30	75.7	46.3	75.1	5.7
10:00	75.3	60.0	82.4	8.3
10:30	74.7	68.7	96.7	8.7
11:00	67.0	60.3	103.5	10.7
11:30	65.3	61.7	106.2	12.7
12:00 noon	62.7	58.0	102.1	10.3
12:30 P.M.	63.7	59.7	115.2	11.3
1:00	47.7	46.3	127.2	16.3
1:30	71.3	62.0	105.0	9.3
2:00	72.3	60.3	66.2	7.0
2:30	65.3	59.3	56.4	6.3
3:00	47.7	12.3	22.3	2.0
3:30	41.3	7.7	15.8	0.0
4:00	22.0	4.7	14.2	0.0
4:30	16.3	2.3	11.2	0.0
5:00	15.3	0.3	5.0	0.0
5:30	6.0	0.0	0.0	0.0
6:00	5.0	0.0	0.0	0.0
Totals	1,164.30	884.7	81.0[b]	136.3

[a]Voluntarily disconnected by caller
[b]Weighted average

Capacity = 1164 calls per day

This is throughput!

Exhibit 3

Type of Call	Frequency (%)	Standard Time (secs)
Order - Standard	60	85
Order - Not Standard	15	120
Status Check	15	220
New Account	5	450
Supply Order	3	125
Miscellaneous	2	120

Avg. Service Time $= 0.6\times85+0.15\times120+\ldots+0.02\times120$
$= 130.7$ secs

$\text{Capacity}_{\text{CSR}}$ = 1 call per 130.7 sec, or
= 27.5 calls/hr

Now what?

Facts:

- ✧ We lose 136.3+ calls per day
- ✧ 1 CSR = 27.5 calls per hour

Conclusion:

- ✧ $\dfrac{136.3 \text{ calls/day}}{27.5 \text{ calls/CSR-hr}} = 5 \text{ CSR-hrs/day}$
- ✧ Therefore, hire 1 CSR. (7.5 hours per day)

Is this a good solution?

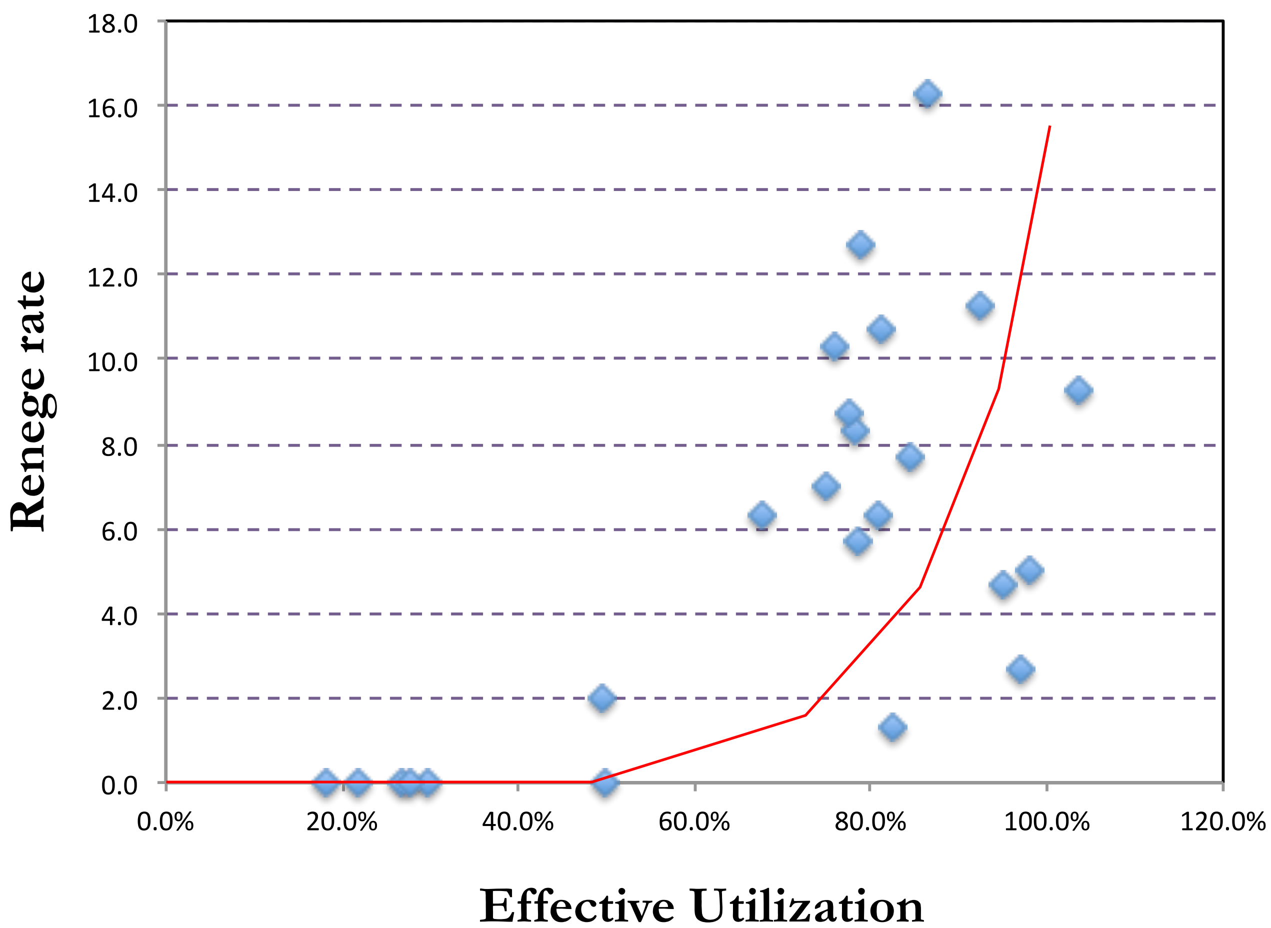
NO: Ignores variability! Ignores scheduling!

Effective Utilization

$$\text{Effective Utilization} = \frac{\lambda_{eff}}{(\#\text{ CSR}) \times (\text{Capacity of a CSR})}$$

Half Hour Ending	Number of Calls	Number Delayed	Average Delay (Seconds)	Number Abandoned	Time Available (min) [Exhibit 4]	#CSR (Av. Time/30)	Total Capacity (27.55/2*#CSRs)	Effective Utilization
6:30 A.M.	22.7	5.7	21.2	1.3	60.0	2.0	27.6	82.4%
7:00	26.7	10.7	28.6	2.7	60.0	2.0	27.6	96.9%
7:30	52.3	41.7	35.7	4.7	120.0	4.0	55.1	94.9%
8:00	54.0	47.7	49.3	5.0	120.0	4.0	55.1	98.0%
8:30	55.7	49.7	61.4	6.3	150.0	5.0	68.9	80.9%
9:00	58.3	59.3	73.8	7.7	150.0	5.0	68.9	84.6%
9:30	75.7	46.3	75.1	5.7	210.0	7.0	96.4	78.5%
10:00	75.3	60.0	82.4	8.3	210.0	7.0	96.4	78.1%
10:30	74.7	68.7	96.7	8.7	210.0	7.0	96.4	77.5%
11:00	67.0	60.3	103.5	10.7	180.0	6.0	82.7	81.1%
11:30	65.3	61.7	106.2	12.7	180.0	6.0	82.7	79.0%
12:00 noon	62.7	58.0	102.1	10.3	180.0	6.0	82.7	75.9%
12:30 P.M.	63.7	59.7	115.2	11.3	150.0	5.0	68.9	92.5%
1:00	47.7	46.3	127.2	16.3	120.0	4.0	55.1	86.6%
1:30	71.3	62.0	105.0	9.3	150.0	5.0	68.9	103.5%
2:00	72.3	60.3	66.2	7.0	210.0	7.0	96.4	75.0%
2:30	65.3	59.3	56.4	6.3	210.0	7.0	96.4	67.7%
3:00	47.7	12.3	22.3	2.0	210.0	7.0	96.4	49.5%
3:30	41.3	7.7	15.8	0.0	180.0	6.0	82.7	50.0%
4:00	22.0	4.7	14.2	0.0	180.0	6.0	82.7	26.6%
4:30	16.3	2.3	11.2	0.0	120.0	4.0	55.1	29.6%
5:00	15.3	0.3	5.0	0.0	120.0	4.0	55.1	27.8%
5:30	6.0	0.0	0.0	0.0	60.0	2.0	27.6	21.8%
6:00	5.0	0.0	0.0	0.0	60.0	2.0	27.6	18.1%

Utilization as a Lever



Determining the "right" number of CSRs

Idea:

- To achieve a small number of lost calls, we need to set a utilization target that balances service level and staffing costs.
- For example, set Target Utilization: $\rho = 65\%$.

Example: Staffing at 12:30

- Let n = number of CSRs required
- Target Utilization = ρ = (Actual Throughput)/Capacity
  - Actual Throughput = (63.7+11.3) calls/hal-hr
  - Capacity = n $\times$ (27.5/2 calls/half-hr)
- Solve: $65\% = (63.7+11.3)\ /(n \times 27.5/2) \approx 8.4$

n = 9 CSRs

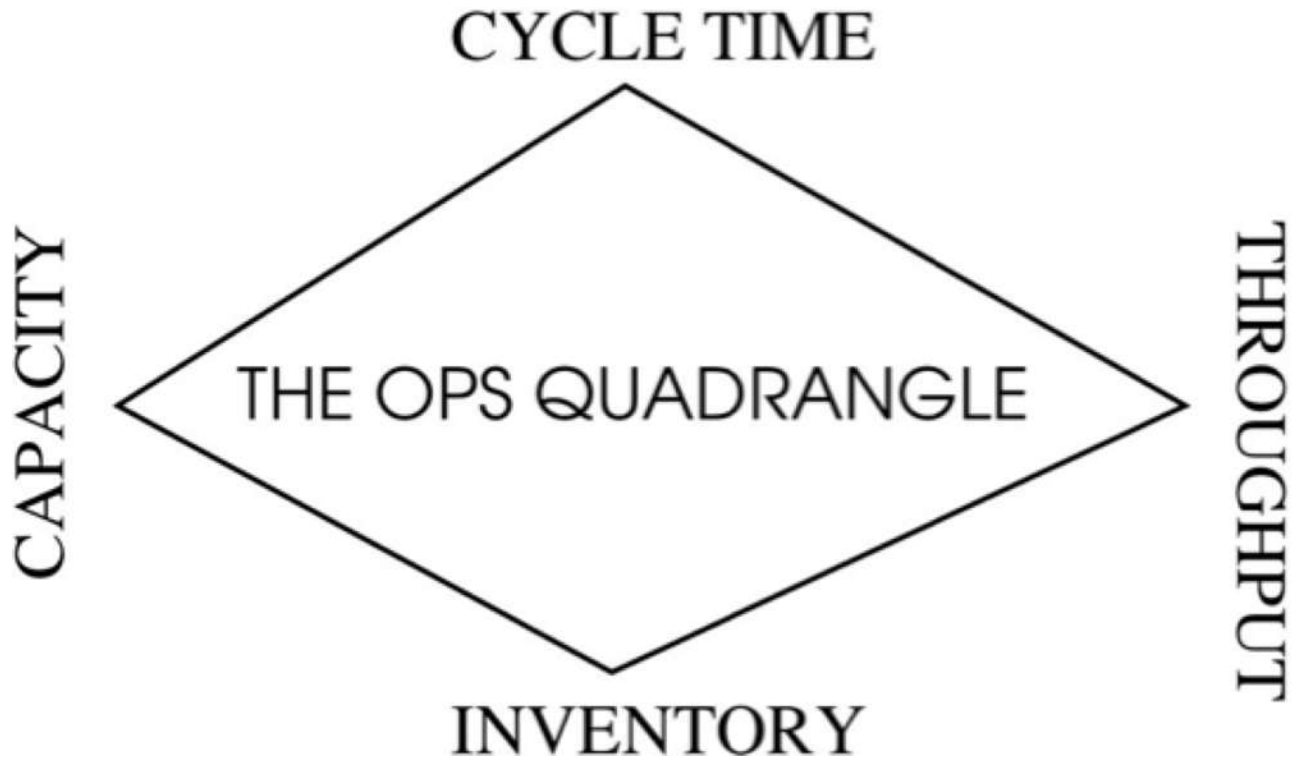
Sof-Optics: Staffing Table Analysis

Half Hour Ending	Number of Calls	Number Delayed	Average Delay (Seconds)	Number Abandoned	Time Available (min) [Exhibit 4]	#CSR Available (Av. Time/30)	#CSR Needed	Effective Utilization	∆CSR Needed
6:30 A.M.	22.7	5.7	21.2	1.3	60.0	2.0	3.0	58.2%	1.0
7:00	26.7	10.7	28.6	2.7	60.0	2.0	4.0	53.5%	2.0
7:30	52.3	41.7	35.7	4.7	120.0	4.0	7.0	59.2%	3.0
8:00	54.0	47.7	49.3	5.0	120.0	4.0	7.0	61.3%	3.0
8:30	55.7	49.7	61.4	6.3	150.0	5.0	7.0	64.4%	2.0
9:00	58.3	59.3	73.8	7.7	150.0	5.0	8.0	60.0%	3.0
9:30	75.7	46.3	75.1	5.7	210.0	7.0	10.0	59.2%	3.0
10:00	75.3	60.0	82.4	8.3	210.0	7.0	10.0	60.8%	3.0
10:30	74.7	68.7	96.7	8.7	210.0	7.0	10.0	60.7%	3.0
11:00	67.0	60.3	103.5	10.7	180.0	6.0	9.0	62.8%	3.0
11:30	65.3	61.7	106.2	12.7	180.0	6.0	9.0	63.0%	3.0
12:00 noon	62.7	58.0	102.1	10.3	180.0	6.0	9.0	59.0%	3.0
12:30 P.M.	63.7	59.7	115.2	11.3	150.0	5.0	9.0	60.6%	4.0
1:00	47.7	46.3	127.2	16.3	120.0	4.0	8.0	58.2%	4.0
1:30	71.3	62.0	105.0	9.3	150.0	5.0	10.0	58.6%	5.0
2:00	72.3	60.3	66.2	7.0	210.0	7.0	9.0	64.1%	2.0
2:30	65.3	59.3	56.4	6.3	210.0	7.0	9.0	57.9%	2.0
3:00	47.7	12.3	22.3	2.0	210.0	7.0	6.0	60.2%	-1.0
3:30	41.3	7.7	15.8	0.0	180.0	6.0	5.0	60.1%	-1.0
4:00	22.0	4.7	14.2	0.0	180.0	6.0	3.0	53.3%	-3.0
4:30	16.3	2.3	11.2	0.0	120.0	4.0	2.0	59.3%	-2.0
5:00	15.3	0.3	5.0	0.0	120.0	4.0	2.0	55.6%	-2.0
5:30	6.0	0.0	0.0	0.0	60.0	2.0	1.0	43.6%	-1.0
6:00	5.0	0.0	0.0	0.0	60.0	2.0	1.0	36.4%	-1.0
								Total	**38.0**

Target Utilization = 0.65
Capacity per CSR = 27.5 calls/hr

Extra CSR hours needed=		**19**
Avg. hourly wage =	**$**	**7.94**
Incremental daily wage costs=	**$**	**150.86**
Incremental annual wage costs=	**$**	**37,715**
Margin lost/call =	**$**	**27.50**
Annual calls abandoned =		**34,075**
Annual margin lost abandoned calls =	**$**	**937,063**
Total annual margin savings =	**$**	**899,348**

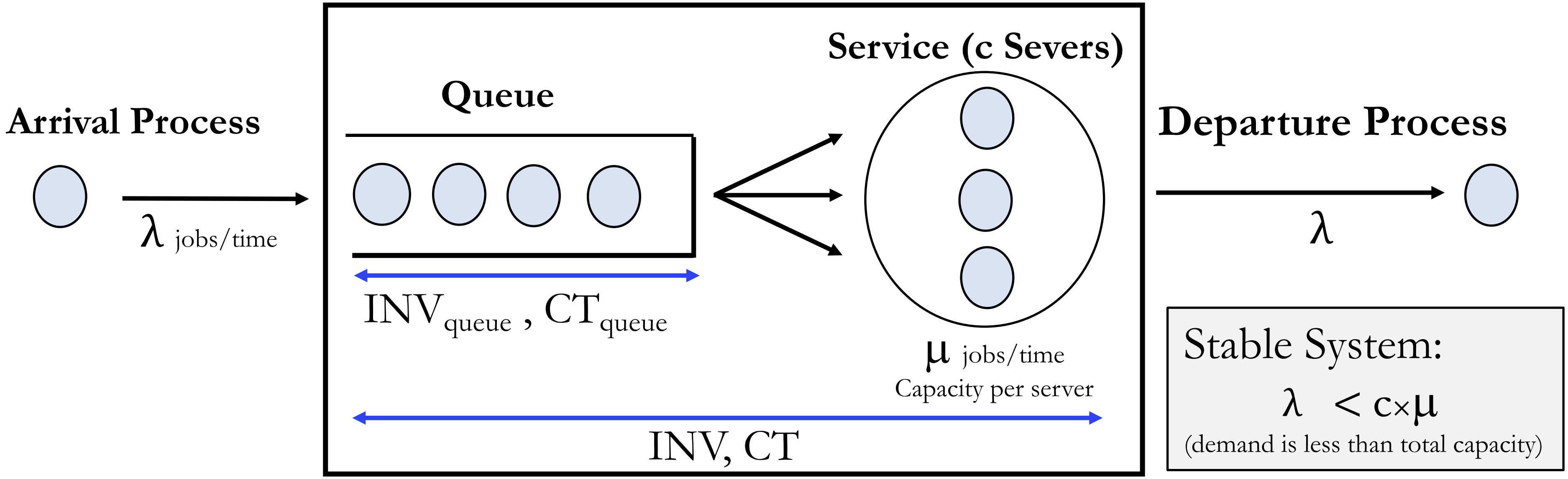
Operations Quadrangle under Uncertainty



New questions:

- What are the effects of variability (random fluctuations) on performance?
- How can we deal with them?

A Simple Business Model



The performance of the system is monitored through

- # Units in the System: $INV = INV_{queue} + INV_{Service}$ & Queue Length: INV_{queue}
- System Time (Flow Time): $CT = CT_{queue} + CT_{Service}$ & Waiting Time: CT_{queue}
- Server Utilization: $\rho = \lambda / (c \times \mu)$

Ops Challenge: In general, arrival and service processes are subject to random fluctuations, which negatively impact the performance of the system.

How should we measure variability?

Two Random Quantities:

t_a = Interarrival times (time between customer arrivals)

t_s = Service times

Mean (Average) Values:

Mean Interarrival Time = $1/\lambda$

Mean Service Time = $1/\mu$

Example:

If λ =10 jobs arrive per hour then the average interarrival time is 1/10 hrs/job.
(i.e., one job every six minutes)

How about variability? Common measure is Standard Deviation: σ

Example: Suppose the Std. Dev. of the service time of call center agent is σ_s = 10 minutes.

Is this a high or low volatility service process?

- If mean service is 60 minutes
- If mean service is 5 minutes

We need a **relative** measure!

Squared coefficient of variation (SCV): $C^2 = \sigma^2/t^2$

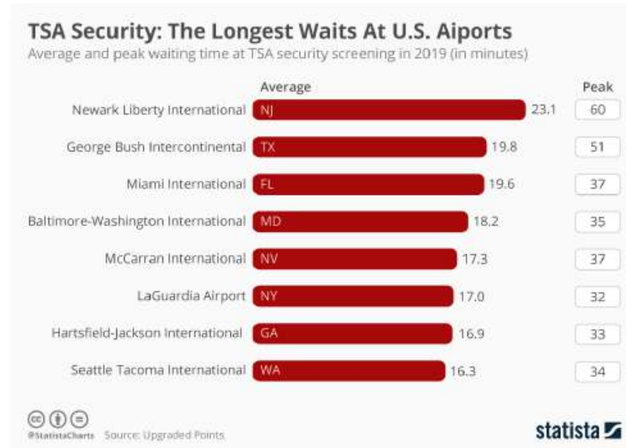
$C^2 < 0.5$	Low relative variability
$0.5 < C^2 \leq 1$	Medium relative variability
$C^2 > 1$	High relative variability

Arrivals: $C_a^2 = \left(\dfrac{\sigma_a}{t_a}\right)^2$

Service: $C_s^2 = \left(\dfrac{\sigma_s}{t_s}\right)^2$

How to Calculate?

Consider a security checkpoint at a local airport staffed by a single officer verifying passengers' travel documents. The following table show a sample of 10 inter-arrival and service times at the checkpoint



Passenger	Inter-arrival (seconds)	Service time (seconds)
1	4	2
2	10	6
3	14	9
4	18	14
5	22	21
6	28	28
7	36	37
8	44	47
9	52	57
10	72	62.5
Mean	$t_a = 30.0$	$t_s = 28.35$
Std. dev.	$\sigma_a = 21.15$	$\sigma_s = 21.70$

Compute:

Throughput: $\lambda = 1/30$ pass/sec $= 2$ pass/min

Capacity: $\mu = 1/28.35$ pass/sec $= 2.12$ pass/min

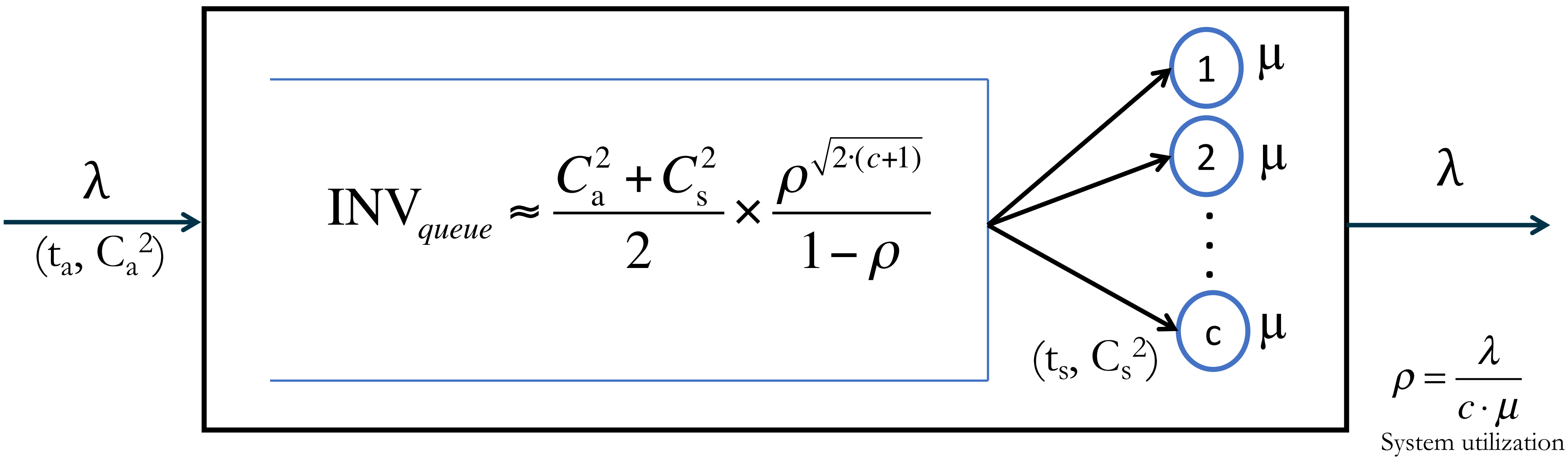
Utilization: $\rho = \lambda / \mu \approx 94.5\%$

Coef. Variation Arrivals: $C_a^2 = (21.15/30.0)^2 \approx 0.497$

Coef. Variation Service: $C_s^2 = (21.70/28.35)^2 \approx 0.585$

The G/G/c Queue

$$\text{Waiting Delays} \propto \text{Process Variability} \times \frac{1}{\text{Capacity Slack}}$$



- Process Variability Effect

$$\frac{C_a^2 + C_s^2}{2}$$

- Capacity/Utilization Effect

$$\frac{\rho^{\sqrt{2\,(c+1)}}}{1-\rho}$$

Little's Law and the G/G/c Queue

$$INV_{queue} \approx \frac{C_a^2 + C_s^2}{2} \times \frac{\rho^{\sqrt{2\cdot(c+1)}}}{1-\rho}$$

Using Little's Law:

$$CT_{queue} = \frac{INV_{queue}}{\lambda}$$

$$CT = CT_{queue} + t_s$$

$$INV = INV_{queue} + \frac{\lambda}{\mu}$$

Special Cases:

- The M/M/c Queue:
  - Poisson Arrivals (Ca=1) and Exponential Service Times (Cs=1)

M/M/c

$$\boxed{INV_{queue} = \frac{\rho^{\sqrt{2(c+1)}}}{1-\rho}}$$

- The M/M/1 Queue:

M/M/1

$$\boxed{INV_{queue} = \frac{\rho^2}{1-\rho}}$$

$$CT = \frac{1}{\underbrace{\mu - \lambda}_{\text{Slack Capacity}}}$$

Example 1

Consider a security checkpoint at a local airport staffed by a single officer verifying passengers' travel documents. The following table show a sample of 10 inter-arrival and service times at the checkpoint.

Passenger	Inter-arrival (seconds)	Service time (seconds)
1	4	2
2	10	6
3	14	9
4	18	14
5	22	21
6	28	28
7	36	37
8	44	47
9	52	57
10	72	62.5
Mean	$t_a = 30.0$	$t_s = 28.35$
Std. dev.	$\sigma_a = 21.15$	$\sigma_s = 21.70$

Throughput: $\lambda = 2$ pass/min

Server's Capacity: $\mu = 2.12$ pass/min

Single server's utilization: $\rho = \frac{\lambda}{\mu} = 0.945$

Coef. Variation Arrival: $C_a^2 = 0.497$

Coef. Variation Service: $C_s^2 = 0.585$

1. What is the average queue length?
2. What is the average cycle time experienced by a passenger?
3. What is the impact on queue length and average cycle time of adding a second security officer?
4. Suppose that instead of adding another human officer, the airport installs a touchless biometric checkpoint. The service time of this system is essentially constant (i.e., no variability) with a mean of 20 seconds. What would be the impact on queue length and average cycle time?

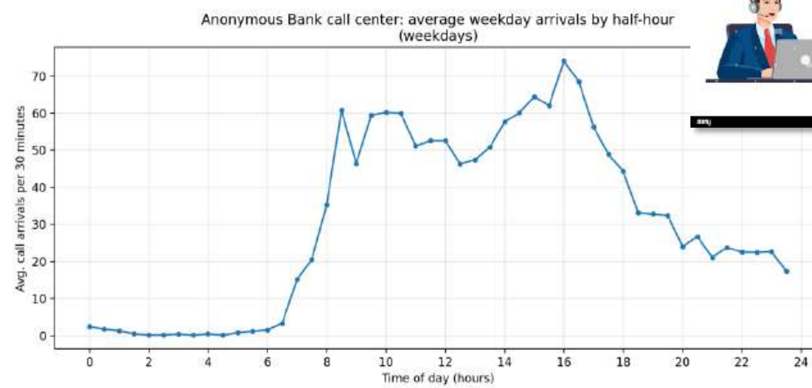
Example 2

Call Center

- 1,600 calls per day on an average
- Each call takes 3 minutes on an average
- Each agent works for 8 hours every day.

Additional Data: Squared coefficient of variation (SCV):

- Arrival = 0.102
- Service = 1.638

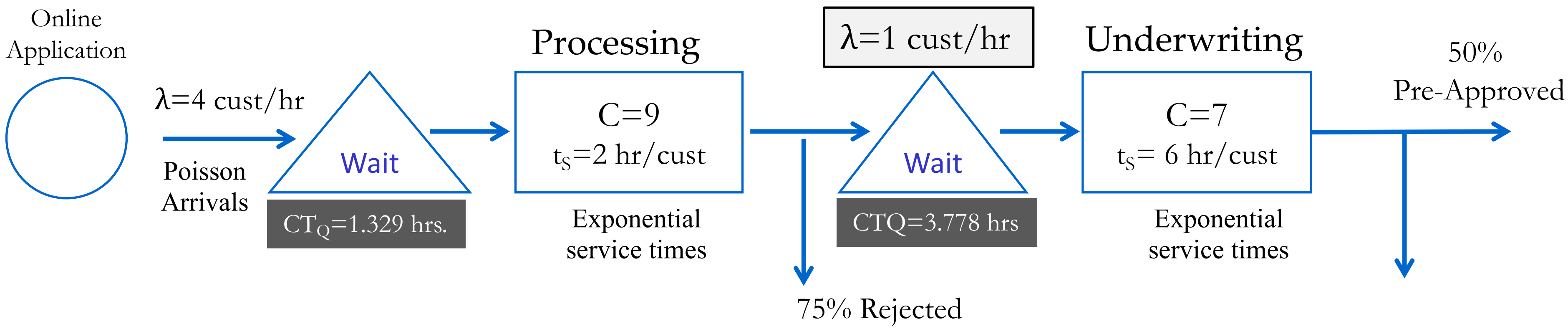


Source: Guedj and Mandelbaum [Link]

Question: What is the optimal number of employees to have?

Example 3

Consider the following (simplified) loan processing office



a) What is the average flow time of a pre-approved loan?

Processing (using M/M/C Excel file)

Number of servers	c	9
Throughput (arrival rate per unit time)	λ	4
Mean Service Time per unit	Ts	2
Service rate per server per unit time	$\mu = 1/T_s$	0.5

Special Case: M/M/c	ρ	0.88889
Probability of empty system	P0	0.0002
Number of units in the system	INV	13.3147
Average time spent in the system	CT	3.32868
Average Waiting Time	CT_Q	1.32868
Length of the queue	INV_Q	5.31473

Underwriting (using M/M/C Excel file)

Number of servers	c	7
Throughput (arrival rate per unit time)	λ	1
Mean Service Time per unit	Ts	6
Service rate per server per unit time	$\mu = 1/T_s$	0.16667

Special Case: M/M/c	ρ	0.85714
Probability of empty system	P0	0.00158
Number of units in the system	INV	9.77843
Average time spent in the system	CT	9.77843
Average Waiting Time	CT_Q	3.77843
Length of the queue	INV_Q	3.77843

Flow Time = 3.3287 + 9.7784 = 13.1 hrs. per customer

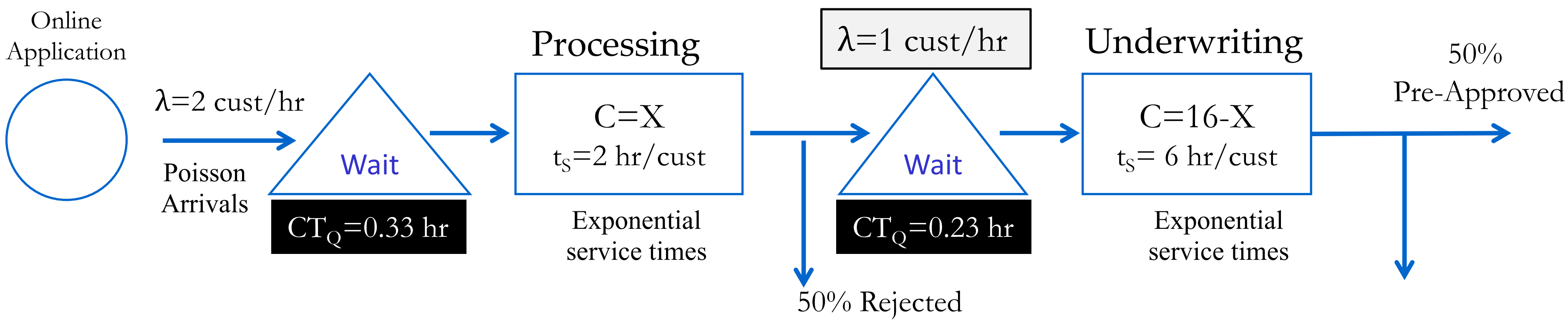
CT Performance = 8 / 13.1 = 61.0%

b) Do you see any inefficiencies in the process?

Example 3 (cont'd)

Suppose you can improve customer screening in the online loan application process. As a result, fewer customers will apply, but a smaller percentage of these applications will be rejected after processing. Specifically, let λ = 2 cust/hr and only 50% of them get rejected after processing.

c) What is the new average cycle time for a pre-approved loan?
Assume that staffing can be adjusted between Processing and Underwriting.



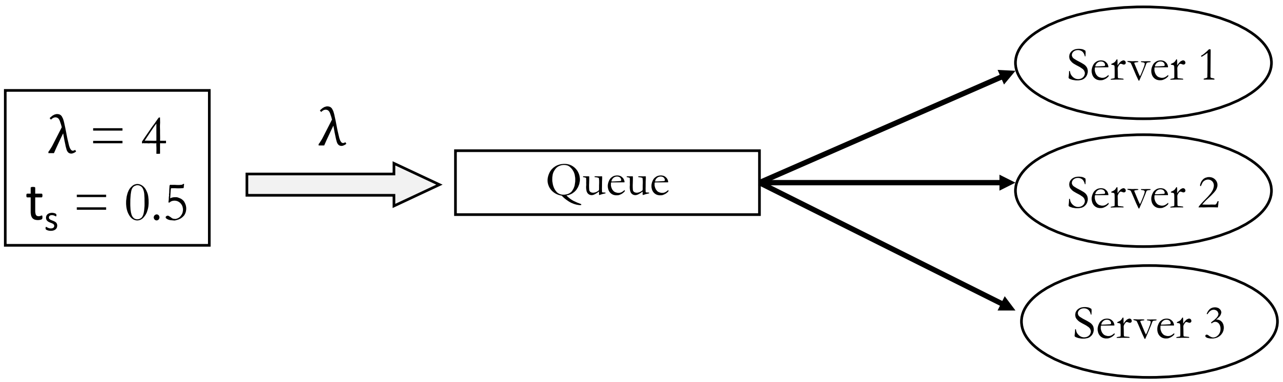
X = number of employees working in Processing (decision variable)

	Process Cycle Time (hrs/cust)		
X	Processing	Underwriting	Total
5	3.15	6.11	9.27
6	2.33	6.23	8.56
7	2.12	6.49	8.61
8	2.05	7.18	9.23
9	2.02	9.78	11.80

CT Performance = 8 / 8.56 = 93.5%

System Design

System 1: Single Queue & Three Identical Server System (M/M/3)

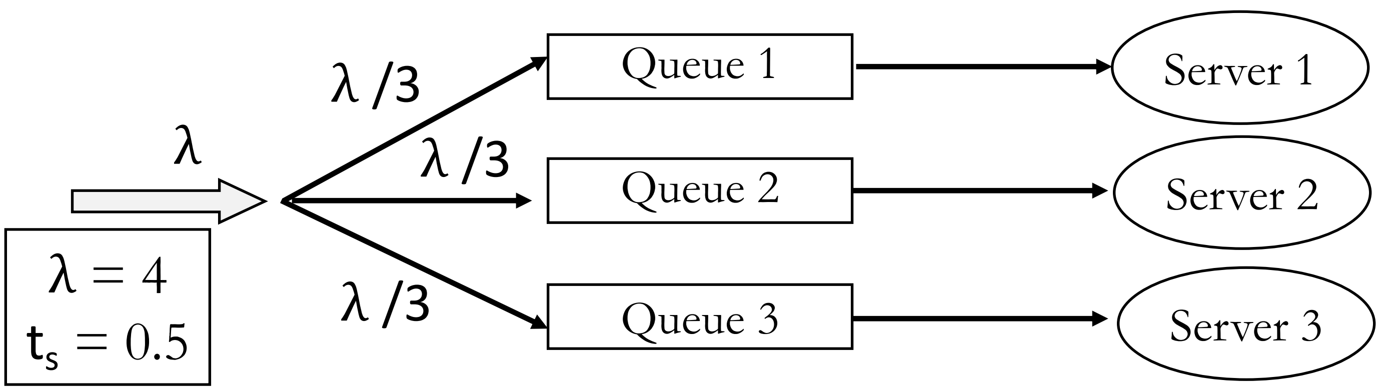


Single (Serpentine) Line

CT = 0.72
INV = 2.89
INV_Q = 0.89

Multiple Lines

System 2: Three Single-Queue-Single-Server System (three M/M/1 in parallel)



CT = 1.5
INV = 6.0
INV_Q = 1.33

Available on YouTube [Link]

Resource Pooling

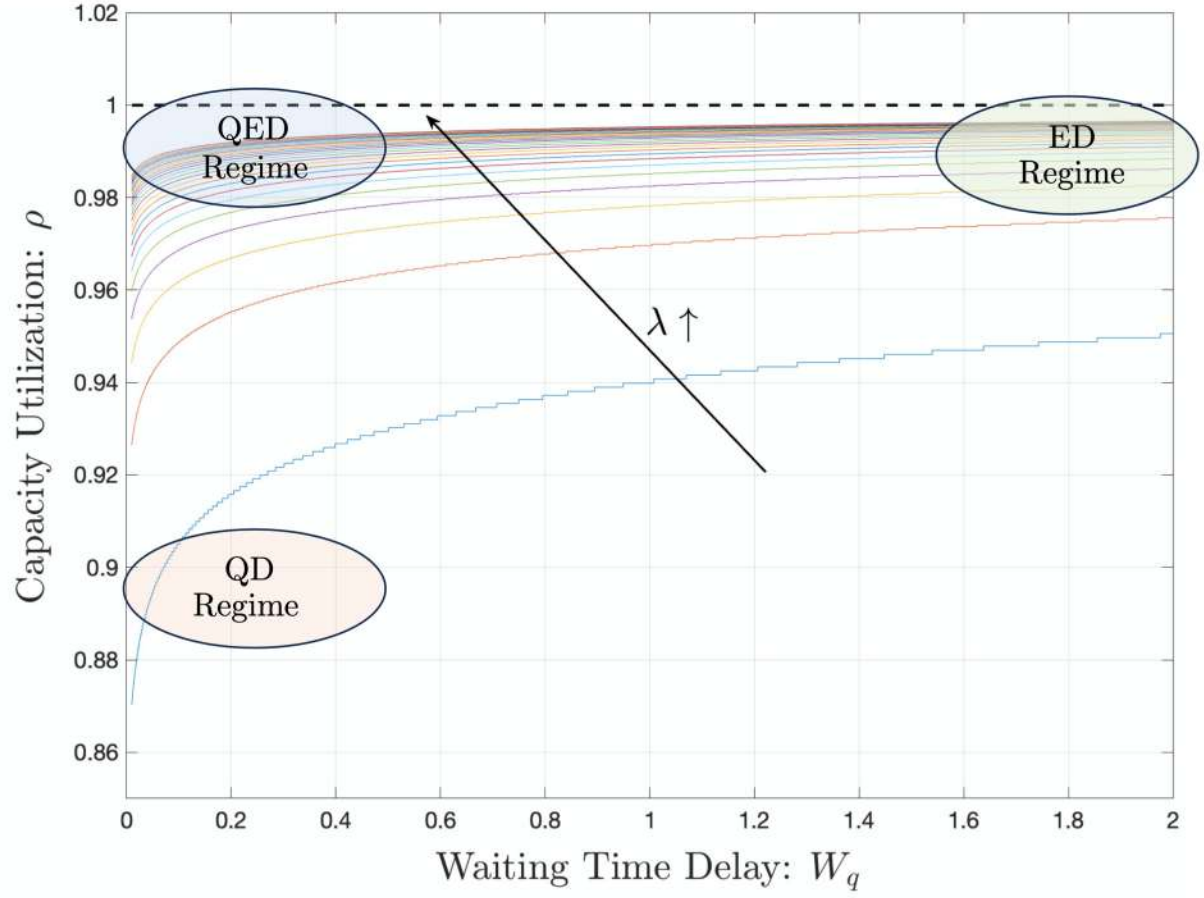
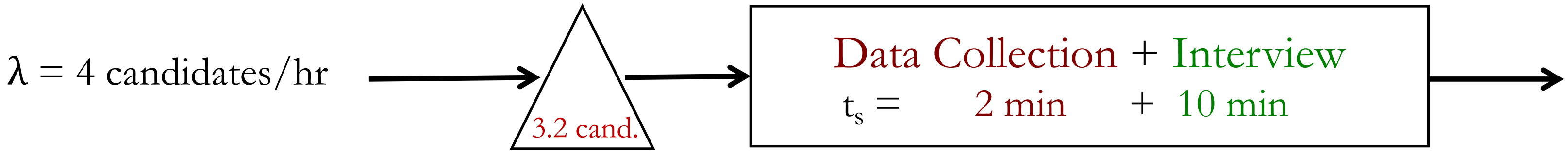


Figure 1: Service System Operating Regimes: Quality-Driven (QD), Efficiency-Driven (ED), and Quality-and-Efficiency-Driven (QED). The QED regime arises under heavy traffic, where both short delays and high utilization are simultaneously achieved through capacity pooling.

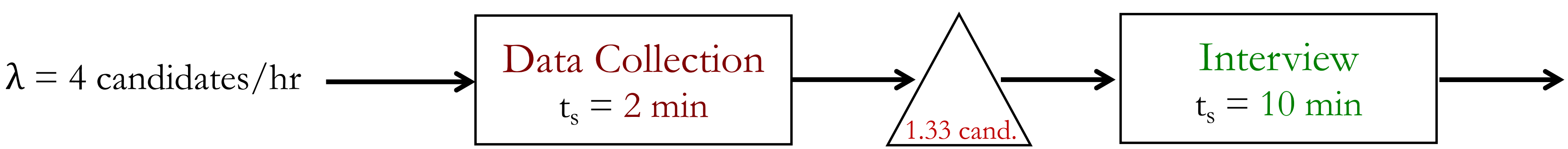
System Design: Self-Service

System 1: Current Interview Process (single server)



CT=1/(5-4)=1hr

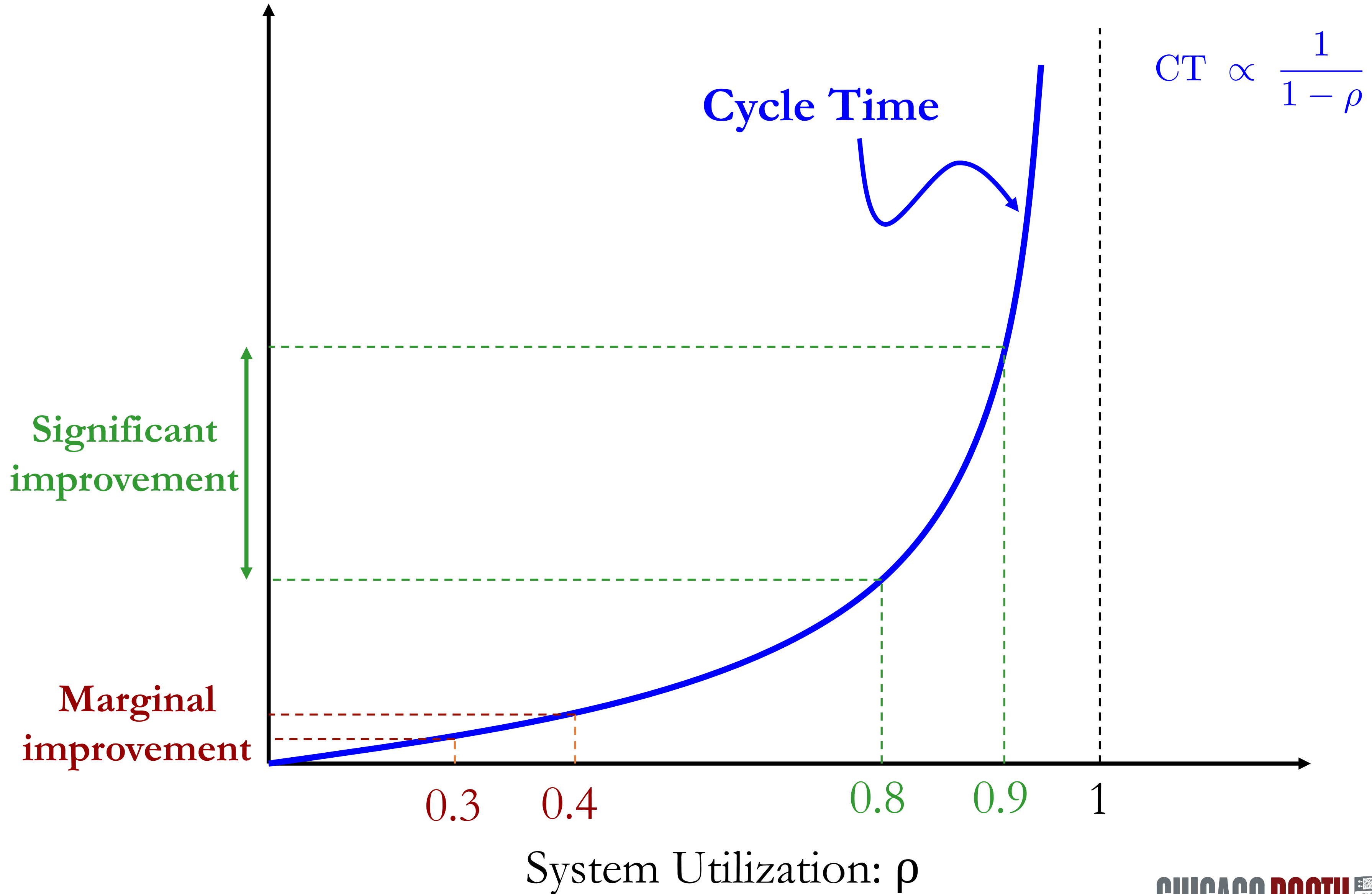
System 2: Interview Process reengineering (Data Collection is self-served)



CT_{Data}= 2 min

$CT_{Interview}$= 1/(6-4)=0.5hrs
= 30 min

Utilization and Variability Drive Congestion



Psychology of Waiting Lines

The Law of Service: Satisfaction = Perception - Expectation

The Principles of Waiting by David Maister

- ✧Unoccupied Time Feels Longer than Occupied Time
- ✧Preprocess Waits Feel Longer than In-Process Waits
- ✧Anxiety Makes Wait Seem Longer
- ✧Uncertain Waits are Longer Than Known, Finite Waits
- ✧Unexplained Waits are Longer Than Explained Waits
- ✧Unfair Waits are Longer than Equitable Waits
- ✧The More Valuable the Service, the Longer the Customer will Wait
- ✧Solo Waits Feel Longer than Group Waits

Waiting Line Experts

Capacity Expansion
Manage Waiting Line Experience
Virtual Queues

Key Ideas

- Waiting lines form due to
  - Variability & the Imbalance between supply and demand

- How to reduce variability
  1. Standardize and Pre-process the service
  2. Train your employees and your customers

- How to match supply and demand
  1. **Supply Management:**
     - Select the right performance measure: cost, utilization, waiting times, queue lengths, etc.
     - Select the best system configuration. Take into account the effects of pooling capacity (single vs. multiple lines)
     - Solve the Staffing problem. Take into account the tradeoff between specialized and flexible capacity.
     - Keep in mind that over-emphasis on utilization could significantly hurt overall performance of the business.
  2. **Demand Management:**
     - Use marketing strategies (pricing, promotions) to smooth arrival pattern.
     - Keep in mind the psychological aspect of waiting

After Class Activities

- REVIEW SESSION 2

- INDIVIDUAL:

  TAKEAWAY #4:

  Contribute a second individual idea for final group project, similar to takeaway 1. (1 paragraph)

- GROUP:

  CASE: TOYOTA MOTOR MANUFACTURING

  Quality as an operational conduit for process improvement